

Day 3: EC2 Instance Families, Access Control, Cloud Storage, and Command-Line Copying

Introduction

Cloud computing has revolutionized the way organizations manage computing resources. Amazon Web Services (AWS) provides various cloud services that help businesses build scalable, secure, and cost-effective applications. Among these services, Amazon EC2 provides virtual servers, cloud storage offers reliable data storage, access control ensures security, and command-line tools simplify file management and administration. Together, these technologies form the foundation of modern cloud infrastructure.

1. EC2 Instance Families

Amazon Elastic Compute Cloud (EC2) provides virtual servers called **instances**. AWS offers different **instance families**, each optimized for specific workloads. Choosing the appropriate instance family improves application performance and minimizes operational costs.

1.1 General Purpose Instances

General Purpose instances provide a balanced combination of CPU, memory, and networking resources. They are suitable for applications that require a balance of computing power and memory.

Applications:

- Web servers
- Small and medium databases
- Development and testing
- Enterprise applications

Examples: T3, T4g, M5, M6i

Advantages

- Cost-effective
- Balanced performance
- Suitable for most business applications

1.2 Compute Optimized Instances

Compute Optimized instances provide high-performance processors designed for CPU-intensive workloads.

Applications:

- Scientific simulations
- Gaming servers
- Machine learning inference

- Batch processing
- High-performance computing (HPC)

Examples: C5, C6i

Advantages

- High CPU performance
- Faster processing
- Reduced execution time

1.3 Memory Optimized Instances

Memory Optimized instances are designed for applications requiring large amounts of RAM.

Applications:

- SAP applications
- In-memory databases
- Real-time analytics
- Big data processing

Examples: R5, R6i, X2idn

Advantages

- High memory capacity
- Better database performance
- Faster data processing

1.4 Storage Optimized Instances

Storage Optimized instances provide high-speed local storage with low latency.

Applications:

- NoSQL databases
- Data warehousing
- Log processing
- Data analytics

Examples: I3, I4i, D3

Advantages

- High input/output performance

- Low storage latency
- Reliable data access

1.5 Accelerated Computing Instances

Accelerated Computing instances use GPUs or specialized hardware accelerators for demanding computational tasks.

Applications:

- Artificial Intelligence
- Machine Learning
- Video rendering
- Scientific research

Examples: P4, G5, F1

Advantages

- High computational power
- Faster AI model training
- Excellent graphics performance

2. Access Control

Access Control is a security mechanism that determines who can access cloud resources and what actions they are allowed to perform. Proper access control protects sensitive information from unauthorized users.

AWS Identity and Access Management (IAM)

IAM enables administrators to create users, groups, and roles and assign permissions using policies.

IAM Components

Users

Represents an individual person or application.

Groups

A collection of users sharing the same permissions.

Roles

Temporary identities assigned to AWS services or users.

Policies

JSON documents defining permissions.

Multi-Factor Authentication (MFA)

MFA enhances security by requiring two forms of authentication:

- Password
- Mobile authentication code or hardware token

This significantly reduces unauthorized access.

Security Groups

Security Groups act as virtual firewalls for EC2 instances.

They control:

- Inbound traffic
- Outbound traffic

For example:

- Allow SSH on Port 22
- Allow HTTP on Port 80
- Allow HTTPS on Port 443

Network Access Control List (NACL)

Network ACLs operate at the subnet level and provide an additional layer of network security.

Unlike Security Groups, Network ACLs support both Allow and Deny rules.

Key Pair

A Key Pair consists of:

- Public Key
- Private Key

It enables secure SSH login to Linux EC2 instances.

3. Cloud Storage

Cloud Storage allows users to store data on remote servers accessed through the Internet.

AWS provides different storage services depending on the application requirements.

Amazon S3 (Simple Storage Service)

Amazon S3 is an object storage service that stores files as objects inside buckets.

Features

- Unlimited storage

- High durability
- Automatic scalability
- Versioning support
- Lifecycle management
- Encryption

Applications

- Image storage
- Video storage
- Website hosting
- Backup and recovery
- Big data analytics

Amazon EBS (Elastic Block Store)

Amazon EBS provides block-level storage for EC2 instances.

Features

- Persistent storage
- High performance
- Snapshot support
- Encryption

Applications

- Operating system disks
- Databases
- Enterprise applications

Amazon EFS (Elastic File System)

Amazon EFS is a scalable shared file storage system.

Applications

- Shared application storage
- Content management systems
- Web servers
- Analytics

Advantages of Cloud Storage

- High availability
- Automatic backup
- Scalability
- Cost efficiency
- Global accessibility
- Disaster recovery

4. Command-Line Copying

Command-line tools help administrators transfer files efficiently between local computers, EC2 instances, and cloud storage.

cp Command

The **cp** command copies files and directories within Linux.

Syntax

cp source destination

Example

cp report.txt Backup/

scp Command

The **Secure Copy (scp)** command transfers files securely using SSH.

Copy Local File to EC2

scp -i mykey.pem report.pdf ec2-user@54.20.10.15:/home/ec2-user/

Copy File from EC2 to Local System

scp -i mykey.pem ec2-user@54.20.10.15:/home/ec2-user/report.pdf .

Advantages

- Secure communication
- Encrypted data transfer
- Simple to use

AWS CLI Copy Command

AWS CLI allows interaction with AWS services directly from the command line.

Upload File to Amazon S3

```
aws s3 cp report.pdf s3://my-bucket/
```

Download File from Amazon S3

```
aws s3 cp s3://my-bucket/report.pdf .
```

Copy One Bucket to Another

```
aws s3 cp s3://bucket1/file.txt s3://bucket2/
```

Sync Local Folder with S3

```
aws s3 sync Documents/ s3://my-bucket/
```

Benefits

- Fast file transfer
- Automation support
- Easy backup
- Efficient cloud management

Conclusion

Amazon EC2 instance families provide different computing options for various workloads, enabling organizations to optimize cost and performance. AWS Access Control mechanisms such as IAM, Security Groups, NACLs, MFA, and Key Pairs protect cloud resources from unauthorized access. Cloud storage services like Amazon S3, EBS, and EFS provide scalable, secure, and highly available storage solutions. Command-line tools such as **cp**, **scp**, and **aws s3 cp** simplify file management and secure data transfer between local systems, EC2 instances, and cloud storage. Together, these services form the core of AWS cloud computing and support modern, scalable, and secure applications.